THE SOLVAY PROCESS 15

Sodium carbonate (Na_2CO_3, 'washing soda') is a very important industrial chemical. It is a white powder and is soluble in water. **Hard water** is water that contains a lot of magnesium or calcium salts (and therefore doesn't lather soap well). Sodium carbonate is used to soften hard water and in making soap and glass. The Solvay (ammonia–soda) process was invented by Ernest Solvay (1838–1922) in the mid 1860s. It is a multistage process that involves the recycling of most of the by-products involved in the manufacture of the sodium carbonate.

The raw materials for the Solvay process are limestone (calcium carbonate, $CaCO_3$) and sodium chloride (NaCl). Ammonia (NH_3) is also used but this is recycled; only small amounts are added at intervals to replenish that lost during the process. It is a cyclical procedure, but the following sequence will help in understanding the overall process.

A Limestone is heated (often by burning coke, a form of carbon) forming calcium oxide (lime, CaO) and carbon dioxide (CO_2).

1 Write a balanced equation for reaction **A**.

B Ammonia and carbon dioxide are passed into a saturated sodium chloride solution forming ammonium chloride (NH_4Cl) and a precipitate of sodium hydrogen carbonate (sodium bicarbonate, $NaHCO_3$).

2 Write a balanced equation for reaction **B**. (Hint: include water as a reactant).

C The temperature is kept below 15°C to ensure that the sodium hydrogencarbonate (slightly-soluble) doesn't dissolve and can be filtered off.

D The sodium hydrogencarbonate is heated to form sodium carbonate, carbon dioxide and water.

3 Write a balanced equation for reaction **D**.

E The carbon dioxide produced is recycled.

F The calcium oxide is added to water forming calcium hydroxide (slaked lime, $Ca(OH)_2$) solution.

4 Write a balanced equation for reaction **F**.

G The ammonium chloride solution is treated with calcium hydroxide solution, forming ammonia, water and calcium chloride ($CaCl_2$).

5 Write a balanced equation for reaction **G**.

H The ammonia is recycled.

I Calcium chloride is a by-product.

J The overall reaction is sodium chloride solution and calcium carbonate solid react to form sodium carbonate solid and calcium chloride solution.

6 Write a balanced equation for reaction **J**.

Figure 1 shows a simplified version of the major reactions in the Solvay process.

7 Copy the diagram but replace each name with the relevant chemical formula.

8 Give the names of the raw materials used as inputs for the Solvay process.

The Solvay process is not without its drawbacks. The process generates large quantities of heat that have to be dealt with. Ammonia is an air pollutant and must not be allowed to escape. There is not a great demand for calcium chloride.

9 Give three uses of sodium carbonate.

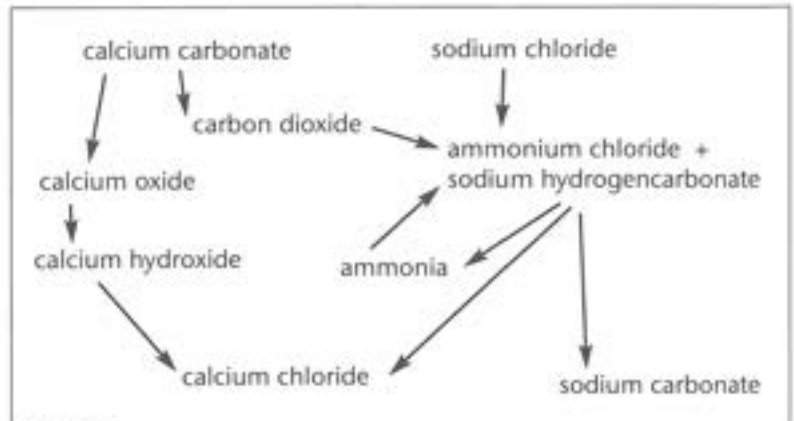


Figure 1

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